RESTRICTION OF SCALARS IS STABLE UNDER BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. 'follows immediately from the definitions' retained verbatim; definitional unpacking recorded in the roadmap as the reviewer's checklist.

Allowed standard prerequisites (editorial).

- Sheaf property of Res. lemma-restriction-of-scalars-sheaf (same chapter, named). Provides that both sides are sheaves.

Source reference. Original lemma 05YB: sources/stacks-criteria/pinned-original-source.tex, source lines 1752–1752. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $X \to Z \to B$ and $B' \to B$ be morphisms of algebraic spaces over S. Set $Z' = B' \times_B Z$ and $X' = B' \times_B X$. Then*

$$Res_{Z'/B'}(X') = B' \times_B Res_{Z/B}(X)$$

in $Sh((Sch/S)_{fppf})$.

Proof. The equality as functors follows immediately from the definitions. The equality as sheaves follows from this because both sides are sheaves according to Lemma 05YB and the fact that a fibre product of sheaves is the same as the corresponding fibre product of pre-sheaves (i.e., functors). □

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